Understanding & Managing Domains

Essential Guide for System Administrators

Domain Registration Process

- **Choose registrar:** Namecheap, GoDaddy, Cloudflare, Gandi
- **Verify availability:** Use WHOIS lookup tools
- **Select TLD:** .com, .org, .net, country codes
- **Configure nameservers:** Point to DNS provider
- **Enable auto-renewal:** Prevent accidental expiration
- **Enable domain lock:** Prevent unauthorized transfers

Essential DNS Records

Record	Purpose	Example
A	IPv4 address	@ → 192.168.1.100
AAAA	IPv6 address	@ → 2001:db8::1
CNAME	Alias to another domain	www → example.com
MX	Mail server	@ → mail.example.com
TXT	Text records (SPF, DKIM)	@ → "v=spf1 include:_spf.google.com ~all"
NS	Nameserver delegation	sub → ns1.subdomain.com

Security Best Practices

- **Enable 2FA:** On registrar and DNS provider accounts
- **Registry lock:** Contact registrar for high-value domains
- **WHOIS privacy:** Hide personal information
- **DNSSEC:** Enable DNS Security Extensions
- **CAA records:** Specify authorized Certificate Authorities

Monitoring & Alerts

- **Expiration monitoring:** Set alerts 90, 30, 7 days before
- **DNS monitoring:** UptimeRobot, Pingdom, StatusCake
- **SSL certificate expiry:** Monitor 30 days before
- **WHOIS changes:** Monitor for unauthorized modifications

Understanding Top-Level Domains (TLDs)

TLD Type	Examples	Use Case
Generic (gTLD)	.com, .org, .net	Commercial, organizations, networks
Country Code (ccTLD)	.us, .uk, .ca, .de	Geographic/national presence
Sponsored (sTLD)	.edu, .gov, .mil	Restricted to specific communities
New gTLD	.tech, .cloud, .app	Industry-specific or branded

Domain Hierarchy Structure

Fully Qualified Domain Name (FQDN) Components:

- **Root domain:** example.com (apex domain)
- **Subdomain:** www.example.com, mail.example.com
- **Third-level:** dev.api.example.com
- **Wildcard domains:** *.example.com (matches any subdomain)

Domain Management Responsibilities

- **Registration maintenance:** Renewals, contact updates
- **DNS configuration:** Records management, nameserver setup
- **Security oversight:** Access control, monitoring changes
- **Service coordination:** Web, email, application hosting
- **Compliance:** Legal requirements, trademark protection
- **Performance optimization:** CDN integration, load balancing

Choosing the Right Registrar

Key factors to consider:

- **Pricing:** Registration, renewal, and transfer costs
- **Features:** DNS management, WHOIS privacy, API access
- **Support:** 24/7 availability, technical expertise
- **Security:** 2FA, domain locking, fraud protection
- **Reliability:** Uptime, infrastructure redundancy
- **Integration:** Works with existing tools and workflows